Singha Junchan

web portfolio

Cloud Security · Identity & Access (IAM) · AI Security · Governance & Risk
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Intro

In 2022 I placed 1st in Thailand at the national programming olympiad (TOI18), after more than 1,000 hours of practice. Now I work on security that helps a business move forward with confidence. I build the technology, lead the people and never lose sight of the business: eleven security research projects (one Best Paper Award), tech lead for Thammasat University's incident-reporting system, president of a student committee, founder of two coaching platforms, and AI agents for third-party risk at Allianz Ayudhya. I learn fast, adapt when plans change and find the creative way through a problem. Internships from Jan 2027, full-time from May 2027.

Education

B.Eng. in Computer Engineering, minor in Cloud Computing & Cybersecurity — Aug 2023 – May 2027 (expected)
Sirindhorn International Institute of Technology (SIIT), Thammasat University — *GPA 3.67 / 4.00 · full scholarship*

Experience

Information Security Intern – *Allianz Ayudhya, Bangkok* — Jun – Jul 2026

- Learned how third-party risk management works with ISO 27001, NIST and internal policies; mentored by the CISO.
- Built two AI agents that support people's judgement: one drafts due-diligence answers with evidence, a reason and a confidence score, finds gaps and suggests fixes, and a person checks every answer; the other teaches business owners the assessment step by step and warns when a step is missed. Tested both with the Information Security team on real, anonymized cases, then presented them.

Tech Lead – *TU Alert, built for Thammasat University, launching soon* — 2026 – Present

- Lead a two-student team hired by the university to build a LINE incident-reporting system that routes each report to the office that owns it; design the architecture and the security, and write code.
- Security by design: second-step staff sign-in, data retention and erasure, Semgrep and gitleaks in CI (TypeScript, Bun, Next.js).

Founder & Lead Instructor – *TOI Tutor and ABIG Center, online* — 2022 – Present

- Founded two platforms that coach students for Thailand's programming olympiads (TOI, POSN): 500+ hours of teaching, hundreds of students and self-written courses (about 270 pages, 128 recorded lessons), built on 1,000+ hours of competitive C++ coding.

R&D Intern, AI Research Team – *Amity Solutions, Bangkok* — Jul – Aug 2023

- Built RAG chatbots on a knowledge base, turned recent LLM papers into prototypes, tried the newest GPT and Claude models.

Research

Eleven projects with Prof. Somchart Fugkeaw (SIIT) on one question: how to keep real cloud, IoT and AI systems secure without slowing them down. First author on ten; 2 accepted, 3 submitted, 6 in preparation.

FEDAC – *zero-trust access for cloud microservices, checked locally* — First author · submitted to IEEE TDSC

- A control hub verifies each caller once and issues a short-lived capability that a sidecar enforces locally; risk-adaptive sign-in up to a Groth16 zero-knowledge proof, blockchain-anchored audit (Go, Rust, AWS EKS).

U-HAP – *one policy language for Kubernetes, compiled* — Best Paper · IEEE JCSSE 2026 · second author

- Co-developed an authorization webhook where teams write ACL, RBAC, ABAC and deny rules in one language, compiled into indexes for near-constant-time decisions; validated against Open Policy Agent; presented it at JCSSE.

PQ-SCVA-MCK – *post-quantum authorization across Kubernetes clusters* — First author · submitted

- Each cluster verifies every decision against the full set of committed allow and deny rules, signed with post-quantum cryptography, with revocation across domains; a capability chain keeps each request's check light (Go, Hyperledger Fabric, Amazon EKS).

ZKAudit, ZK-ALBA and PQZ-Audit – *senior project: verifiable audit of cloud logs* — First author · ZKAudit accepted, KST 2027

- ZKAudit: auditors prove credentials, scope and policy freshness without revealing them (replay-safe verifier shards, Merkle commitments; Rust, Amazon EC2). ZK-ALBA (journal, in preparation) ties each proof to one audit, one set of logs and one auditor.
- PQZ-Audit (in preparation): auditors run real checks on encrypted logs (time windows, thresholds, totals) and verify the answer is correct and complete without seeing them; t-of-n approval for privileged audits; fully post-quantum (ML-KEM, ML-DSA).

SWARM-Fog – *self-healing orchestration for industrial IoT* — First author · submitted

- Fog clusters share a map of past experience for trust-aware scheduling; recovery plans are ready before nodes fail.

In preparation: MoCA (AI-agent safety), Fed-RAG-Hunt (federated threat hunting), an encrypted SIEM, freshness-aware cloud scaling.

Hands-on Security

NCSA CTF Thailand Tournament 2026 (Thailand Cyber Top Talent), in a team (web, crypto, forensics, OSINT, mobile) · SOSECURE Road to Cybersecurity 2025 (Linux hardening, web pentesting, SOC) · home lab on Kali Linux · TryHackMe, Hack The Box

Skills

Security & IAM:	Zero Trust, OIDC/JWT, MFA (TOTP), RBAC/ABAC, OPA/Rego, Kubernetes authorization
Governance & Risk:	ISO 27001, NIST, third-party risk management, due-diligence questionnaires, gap analysis
Cloud & AI:	AWS (EKS, EC2), Kubernetes, Docker, Azure basics; LLM agents, RAG, MCP, prompt-injection testing
Offensive & SOC:	Nmap, Wireshark, Burp Suite, Metasploit, OSINT, OWASP Top 10; Splunk, Wazuh, Suricata
Cryptography:	zk-SNARKs (Groth16), Merkle commitments, post-quantum ML-DSA and ML-KEM
Programming:	Python and C++ (main); Go, Rust, TypeScript, C, Java, Bash, SQL
Languages:	Thai (native), English (fluent), Chinese (basic)
Interests:	home-lab security research, Muay Thai, board games, food and travel

Awards, Certifications & Leadership

- **1st in Thailand**, national champion and gold medal: Thailand Olympiad in Informatics (TOI18), 2022.
- **Best Paper Award**, IEEE JCSSE 2026 · 1st Place, Image Recognition Challenge (Osaka Inst. of Technology, SIIT), 2025.
- ISC2 Certified in Cybersecurity (CC), 2025 · CompTIA Security+ (exam planned Dec 2026) · DataCamp Azure, AWS, 2025.
- President, Sub Student Committee 32, SIIT (since 2024): organised 11 events, led First Meet 2024 (about 500 people, 100 staff).

References

Apichai Wongchindakhun (CISO, Allianz Ayudhya) | linkedin.com/in/apichai-wongchindakhun-1839b314
Prof. Somchart Fugkeaw (SIIT, research supervisor) | somchart@siit.tu.ac.th
Pachara (TU Alert co-developer) | contact details on request